

Statement of participation

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has completed the free course including any mandatory tests for:

Data analysis: hypothesis testing

This free 9-hour course looked at the principles of hypothesis testing.

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www.open.edu/openlearn

This statement does not imply the award of credit points nor the conferment of a University Qualification.
This statement confirms that this free course and all mandatory tests were passed by the learner.

Please go to the course on OpenLearn for full details:

<https://www.open.edu/openlearn/science-maths-technology/data-analysis-hypothesis-testing/content-section-0>

COURSE CODE: B126_2

Data analysis: hypothesis testing

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Course summary

Making decisions about the world based on data requires a process that bridges the gap between unstructured data and the decision. Statistical hypothesis testing helps decision-making by formulating beliefs about the world, including people, organisations or other objects, and formally testing these beliefs. In this free course, you will study the principles of hypothesis testing, including the specification of significance levels, as well as one-sided and two-sided tests. Finally, you will learn how to perform a hypothesis test of the mean of a variable, as well as the proportion of individuals in a dataset with a certain characteristic. You will use spreadsheets throughout the course as the central tool used by professionals for simple data management and analysis.

Learning outcomes

By completing this course, the learner should be able to:

- understand the principle of hypothesis testing
- understand the idea of alpha in hypothesis testing
- differentiate between one-tailed and two-tailed tests
- understand hypothesis testing of means and proportions
- report the exact p-value of a test.

Completed study

The learner has completed the following:

Section 1

Two types of hypotheses

Section 2

Testing with data

Section 3

Alpha (α) levels

Section 4

One-tailed vs two-tailed test

Section 5

Mean and z-score ranges

Section 6

P-value

Section 7

Hypothesis testing for population proportions